WORKED SOLUTION

STEP III 2003, Question 10

Mechanics • Variable Acceleration

STEP III 2003, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q10

The key idea

The acceleration $kx\dot{x}$ is an exact derivative with respect to time: it is $\frac{\mathrm{d}}{\mathrm{d}t}(\frac{1}{2}kx^2)$. So we can integrate once straight away and get a first-order equation $\dot{x} = \frac{1}{2}kx^2 + C$, which separates. The sign of the constant C decides everything: a positive constant gives an arctangent (and a solution that blows up), a negative one gives partial fractions and a logarithm (and a solution that settles down).

Solution

We treat the particle as moving on the x-axis with $\ddot{x} = kx\dot{x}$. Since $\frac{\mathrm{d}}{\mathrm{d}t}(x^2) = 2x\dot{x}$, the equation of motion is

$$\frac{\mathrm{d}\dot{x}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\tfrac{1}{2}kx^2\right), \qquad \text{so} \qquad \dot{x} = \tfrac{1}{2}kx^2 + C$$

for a constant C. At $t = 0$ we have $x = d$ and $\dot{x} = U$, so $C = U - \frac{1}{2}kd^2$, and

$$\dot{x} = \tfrac{1}{2}k(x^2 - d^2) + U. \qquad (*)$$

Part (i)

With $U = kd^2$, equation $(*)$ becomes

$$\dot{x} = \tfrac{1}{2}k(x^2 + d^2).$$

The right-hand side is always positive, so we may separate variables:

$$\int_d^x \frac{\mathrm{d}u}{u^2 + d^2} = \int_0^t \tfrac{1}{2}k\,\mathrm{d}s \quad \Longrightarrow \quad \frac{1}{d}\left[\arctan\frac{x}{d} - \arctan 1\right] = \tfrac{1}{2}kt.$$

Since $\arctan 1 = \frac{\pi}{4}$, this gives $\arctan(x/d) = \frac{1}{2}dkt + \frac{\pi}{4}$, so

$$x = d\tan\left(\frac{dkt}{2} + \frac{\pi}{4}\right).$$

This is valid as long as the argument of tan stays below $\frac{\pi}{2}$ (the arctangent takes values in $(-\frac{\pi}{2}, \frac{\pi}{2})$). As t increases from 0, the argument increases from $\frac{\pi}{4}$, and it reaches $\frac{\pi}{2}$ when

$$\frac{dkt}{2} = \frac{\pi}{4}, \qquad \text{that is,} \qquad t = \frac{\pi}{2dk}.$$

Since $\tan\theta \to +\infty$ as $\theta \to \frac{\pi}{2}$ from below, and $d > 0$, we get $x \to \infty$ as $t \to \frac{\pi}{2dk}$, as required. The particle reaches infinity in a finite time.

Part (ii)

Now $U < 0$, so $C = U - \frac{1}{2}kd^2 < 0$. Write $C = -\frac{1}{2}ka^2$ with $a > 0$, that is,

$$a = \sqrt{d^2 - \frac{2U}{k}}, \qquad \text{so that} \qquad \dot{x} = \tfrac{1}{2}k(x^2 - a^2).$$

Because $U < 0$ and $k > 0$, we have $a > d > 0$.

? **Ask yourself** Which partial-fraction logarithm should we use: $\ln\frac{x-a}{x+a}$ or $\ln\frac{a-x}{a+x}$?

It depends on where x lives. The constant solutions $x = \pm a$ are where $\dot{x} = 0$. Initially $x = d$ lies strictly between $-a$ and a, and a solution starting there can't cross the constant solutions $x = \pm a$ (two solutions of a first-order equation like this can't meet), so $-a < x < a$ for all time. In that range $\dot{x} < 0$, so x decreases, and the quantities $a - x$ and $a + x$ are both positive.

Using $\dfrac{1}{x^2 - a^2} = \dfrac{1}{2a}\left(\dfrac{1}{x-a} - \dfrac{1}{x+a}\right)$ and separating variables,

$$\frac{1}{2a}\left[\ln\frac{a-u}{a+u}\right]_d^x = \tfrac{1}{2}kt \quad\Longrightarrow\quad \frac{a-x}{a+x} = A\,\mathrm{e}^{akt}, \qquad \text{where } A = \frac{a-d}{a+d}.$$

Note $0 < A < 1$ because $a > d > 0$. Solving for x,

$$x = a\,\frac{1 - A\mathrm{e}^{akt}}{1 + A\mathrm{e}^{akt}}, \qquad a = \sqrt{d^2 - \frac{2U}{k}}, \quad A = \frac{a-d}{a+d}.$$

This is defined for every $t \geq 0$ (the denominator is positive). Equivalently, since $\ln A = -2\,\mathrm{artanh}(d/a)$, it can be written $x = a\tanh(\mathrm{artanh}(d/a) - \frac{1}{2}akt)$.

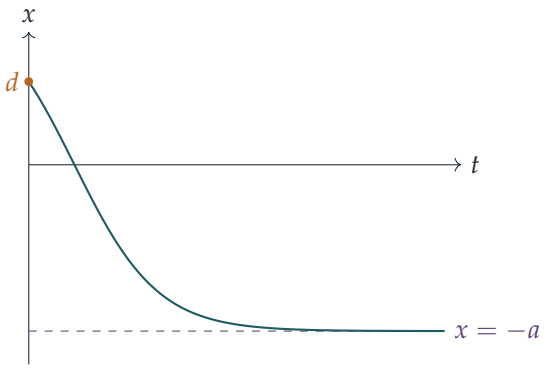
As $t \to \infty$, $A\mathrm{e}^{akt} \to \infty$ because $A > 0$ and $ak > 0$. Dividing top and bottom by $A\mathrm{e}^{akt}$,

$$x = a\,\frac{A^{-1}\mathrm{e}^{-akt} - 1}{A^{-1}\mathrm{e}^{-akt} + 1} \;\longrightarrow\; -a.$$

So x tends to the limit

$$-a = -\sqrt{d^2 - \frac{2U}{k}}.$$

(The limit also involves k; it can't be written using d and U alone, since changing k with d and U fixed changes it.)



The graph (drawn for $d = 1$, $a = 2$, $k = 1$) shows the particle starting at d with negative velocity, passing through the origin and approaching $-a$ ever more slowly.

Checks

- In part (i), $t = 0$ gives $x = d\tan\frac{\pi}{4} = d$, and differentiating gives $\dot{x} = \frac{1}{2}kd^2\sec^2\frac{\pi}{4} = kd^2 = U$.
- In part (ii), $t = 0$ gives $x = a\frac{1-A}{1+A} = d$. As $U \to 0^-$, $a \to d$ and $A \to 0$, so x stays at d: correct, since $x = d$, $\dot{x} = 0$ makes the acceleration zero.
- Numerically, stepping $\ddot{x} = kx\dot{x}$ with a Runge–Kutta method for random k, d and U matched both formulae to better than 10^{-6}. For example, with $k = 1$, $d = 1$, $U = -1.5$ we have $a = 2$, and the numerical x settles at -2.000000.

Common slips

- Using $\ln\frac{x-a}{x+a}$ without absolute values: for $|x| < a$ this is the log of a negative number.
- Taking the positive square root for the limit: x decreases from d, so the limit is $-a$, not $+a$.
- Writing the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}x}$ and getting $v\frac{\mathrm{d}v}{\mathrm{d}x} = kxv$, then dividing by v without noting $v \neq 0$. It works here, but integrating $\frac{\mathrm{d}}{\mathrm{d}t}$ directly avoids the issue.

Beyond STEP

Part (i) is a small example of *finite-time blow-up*: an equation $\dot{x} = f(x)$ with f growing faster than linearly (here quadratically) can send the solution to infinity in finite time, so solutions need not exist for all t. Part (ii) shows the other behaviour: the constant solutions $x = \pm a$ are equilibria, $-a$ is stable and $+a$ unstable, and the phase-line picture predicts the limit before any integration. Both ideas, together with the uniqueness theorem that stops solutions crossing, are covered in Boyce and DiPrima, *Elementary Differential Equations*.

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